An extendable generalized Markov shift with countably infinitely many empty words

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Status. Candidate full solution; likely valid, subject to human review. The construction answers one explicit direction in the concluding remarks of Bissacot–Diaz-Granados–Raszeja [1].

1 Source question

The source paper studies the generalized countable Markov shift X_A, the Gelfand spectrum of the diagonal algebra associated with an irreducible zero-one transition matrix A. Its shift is initially defined only on the open set of non-empty configurations. Section 6, page 35, asks for more general classes on which it extends continuously and observes that all examples constructed there have finitely many empty-stem configurations. It leaves open the construction of examples with infinite E_A.

> of the same problem. More precisely, the maximum value attained among the invariant measures in the right-handed side of (5.17) to calculate $r(a\alpha)$ can actually be taken for invariant measures on a transitive subshift on finite alphabet (that depends on a). And finally, 76 illustrates how the results generalize the finite alphabet case (see Section 6 of [37]).
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> Some new natural questions arise from this paper. We point out that, although Theorem 42 characterizes when the shift map σ admits a continuous extension, and we have an endomorphism α on $\mathcal{D}_A$, where σ is dual to α, the elements $\alpha_0(g)$, $g \in \mathcal{G}_A$, are always bounded operators on $\ell^2(\Sigma_A)$. Since the image of C*-algebras by *-homomorphisms are C*-algebras, maybe it is possible to explore the connections between the C*-algebra generated by $\alpha_0(\mathcal{G}_A)$ with the generalized Markov shifts for the cases where the shift map does not admit a continuous extension.
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> Another possible direction is to investigate more general classes of generalized Markov shifts, where we can extend the shift map continuously. We mention that our classes of examples consist of generalized Markov shifts with finitely many empty-word configurations, so it is open to questions concerning examples for infinite E_A. Moreover, it would be of great interest if we could obtain explicit formulae for $r(a\alpha)$ for general irreducible matrices.

We give such an example with E_A countably infinite. In fact, every empty configuration will be a fixed point of the extended shift.

2 Definitions used from the source

Write $C_x = \{y : A(y,x) = 1\}$ for the support of the column indexed by x. We use the following facts recalled in [1] from [2].

(i) If A is irreducible, the generalized shift X_A is realized as a closed subspace of the configuration cube $\{0,1\}^{\mathbb{F}}$ and is metrizable for a countable alphabet.

(ii) The nonzero accumulation points of the columns (C_x) in $\{0,1\}^{\mathcal{A}}$ are in bijection with the empty-stem configurations $(e,R) \in E_A$; the support R is called the root.

(iii) If non-empty configurations ξ_n converge to (e,R), then their first stem symbols x_n escape every finite subset of the alphabet and $C_{x_n} \to R$ coordinatewise. Conversely, these conditions identify all limit points as (e,R).

(iv) The shift is a local homeomorphism on the open set of non-empty configurations. For a stem of length at least two it deletes the first symbol; for a finite stem of length one it leaves the associated root as an empty configuration.

These statements can also be read directly from the four configuration rules: if the first stem symbol is x, then the value at the inverse coordinate y^{-1} is $A(y,x)$, which is precisely the y-coordinate of C_x.

3 The construction

Let the countable alphabet be

$$\mathcal{A} = \{o\} \sqcup \{b_j : j \geq 1\} \sqcup \{v_{j,k} : j,k \geq 1\}.$$

Think of o as a hub, the b_j as the vertices of a rooted binary tree, and $(v_{j,k})_{k\geq 1}$ as an infinite ray attached to b_j. Define a parent map on $\mathcal{A} \setminus \{o\}$ by

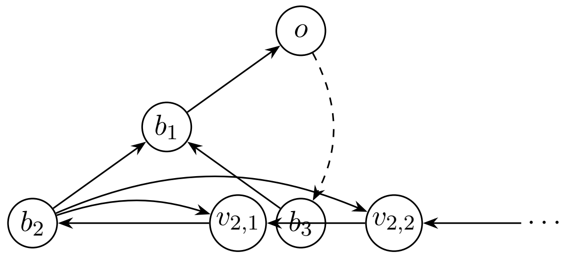
$$\begin{gathered} p(b_1) = o, \qquad p(b_j) = b_{\lfloor j/2 \rfloor} \quad (j \geq 2), \\ p(v_{j,1}) = b_j,\ p(v_{j,k+1}) = v_{j,k} \quad (j,k \geq 1). \end{gathered}$$

Declare the arrows of the Markov graph to be

$$o \longrightarrow x \quad (x \in \mathcal{A}), \qquad x \longrightarrow p(x) \quad (x \neq o), \qquad b_j \longrightarrow v_{j,k} \quad (j,k \geq 1),$$

and no others. After choosing any bijection $\mathcal{A} \simeq \mathbb{N}$, this is a countable zero-one transition matrix A.

The picture below shows the two mechanisms. Arrows toward the hub provide irreducibility; each marker also emits along its own ray.



solid arrows: parent/ray arrows shown locally;
dashed arrow: the hub emits to every symbol.

4 Proof intuition

The hub row is present in every column, so the generalized shift is compact and the zero vector can never be a column limit. Far out on the ray attached to b_j, a column consists of the fixed pair $\{o, b_j\}$ plus one moving coordinate. It therefore converges to $\{o, b_j\}$. If the ray/marker index j itself escapes, even the marker coordinate disappears, leaving $\{o\}$. These are all possible column limits, so they give countably many empty words.

The parent map was chosen so that deleting the first symbol preserves exactly the same limiting root. Along a fixed ray, shifting moves one step toward its marker and preserves $\{o, b_j\}$. Across rays with $j \to \infty$, every possible successor still has class index tending to infinity and hence root $\{o\}$. This is the continuity mechanism.

5 The theorem

Theorem 1. *For the matrix A above:*

(a) A is irreducible and column-finite;

(b) X_A is compact;

(c)
$$E_A = \{(e, R_\infty)\} \cup \{(e, R_j) : j \geq 1\}, \qquad R_\infty = \{o\}, \quad R_j = \{o, b_j\};$$
in particular, E_A is countably infinite;

(d) the shift extends continuously to all of X_A by $\sigma(e, R) = (e, R)$ for every $R \in E_A$.

Consequently the source question about extendable examples with infinite E_A has an affirmative answer.

Proof. For a symbol x, let $p^{-1}(x)$ denote its children under the parent map. The column supports are

$$C_o = \{o, b_1\}, \tag{1}$$

$$C_{b_j} = \{o, b_{2j}, b_{2j+1}, v_{j,1}\}, \tag{2}$$

$$C_{v_{j,k}} = \{o, b_j, v_{j,k+1}\}. \tag{3}$$

Thus every column has at most four entries, proving column-finiteness. From each $x \neq o$, repeated parent arrows reach o: this is clear along a ray and then follows from repeated integer halving in the marker tree. From o there is an arrow to every symbol. Hence every symbol reaches every other symbol, so A is irreducible.

The row indexed by o is identically one. Accordingly, the Exel–Laca projection Q_o acts as the identity on $\ell^2(\Sigma_A)$ and belongs to the diagonal algebra $\mathcal{D}_A$. Therefore $\mathcal{D}_A$ is unital and its Gelfand spectrum X_A is compact.

We next classify the column accumulation points. From (3), for each fixed j,

$$C_{v_{j,k}} \longrightarrow R_j = \{o, b_j\} \qquad (k \to \infty),$$

because the coordinate $v_{j,k+1}$ escapes every finite subset of $\mathcal{A}$. From (2) and (3), whenever $j_n \to \infty$ we have

$$C_{b_{j_n}} \longrightarrow R_\infty = \{o\}, \qquad C_{v_{j_n,k_n}} \longrightarrow R_\infty$$

for arbitrary k_n.

There are no other accumulation points. Indeed, take any sequence of distinct column indices. If it contains infinitely many ray indices from one fixed class j, pass to that subsequence; its depth tends to infinity and its columns tend to R_j. Otherwise, after passing to a subsequence, the marker or ray class indices tend to infinity, and the columns tend to R_∞. The single column C_o causes no accumulation. All the displayed limits are nonzero and distinct. The column-limit/root correspondence now gives exactly the set E_A stated in (c). Notice also that $R_j \to R_\infty$ in the product topology.

It remains to prove continuity after defining every member of E_A to be a fixed point. Since X_A is metrizable, sequential continuity suffices, and the original shift is already continuous on the open set of non-empty configurations. Let non-empty configurations $\xi_n \to (e, R)$ and write x_n for their first stem symbols. Then x_n escapes every finite set and $C_{x_n} \to R$.

First suppose $R = R_j$. The coordinate b_j is eventually present in C_{x_n}. Inspection of (1)–(3) shows that, apart from the one fixed exceptional marker column whose child list contains b_j, the indices must be

$$x_n = v_{j,k_n}, \qquad k_n \to \infty.$$

Discarding finitely many terms removes the exception. A ray symbol is a finite emitter with unique successor $p(v_{j,k_n})$; consequently the stem cannot terminate there. For large n, this successor is v_{j,k_n-1}, whose column again tends to R_j. Therefore every limit point of $\sigma(\xi_n)$ is (e, R_j), and compactness implies $\sigma(\xi_n) \to (e, R_j) = \sigma(e, R_j)$.

Now suppose $R = R_\infty$. The column classification implies that the marker/ray class indices j_n of x_n tend to infinity. If $x_n = v_{j_n,k_n}$, its unique successor is its parent; whether that parent is b_{j_n} or another ray vertex, its column tends to R_∞. If $x_n = b_{j_n}$ and the stem has length at least two, the next symbol is either $p(b_{j_n}) = b_{\lfloor j_n/2 \rfloor}$ or some $v_{j_n,k}$. Since $j_n \to \infty$, the columns of all these possibilities tend to R_∞, uniformly in k on every fixed finite coordinate set.

There is one final finite-stem case. If the stem is the single marker b_{j_n}, its root must be R_{j_n}: among the successors of b_{j_n}, the isolated parent column cannot create a limit, while the columns along its ray have the unique accumulation point R_{j_n}. Deleting the one-letter stem therefore gives

$$\sigma(\xi_n) = (e, R_{j_n}) \longrightarrow (e, R_\infty).$$

Thus $\sigma(\xi_n) \to (e, R_\infty) = \sigma(e, R_\infty)$ in every case. The extension is continuous at all empty configurations and hence on all of X_A. □

6 Verification and limitations

A reusable finite-window script accompanies this packet. It checks the three column formulas, maximum column size four, sampled parent paths to the hub, and stabilization to R_j and R_∞ on finite coordinate windows. It passes, but these computations are only sanity checks; the infinite classification and continuity are proved above.

The result addresses the source's existence question for infinite E_A. It does not provide an explicit closed formula for the spectral radius $r(a\alpha)$ for every irreducible matrix, nor does it characterize all extendable matrices.

7 Novelty search and review recommendation

A bounded current web/arXiv search on 13 August 2026 used the exact source title and the phrases "infinite E_A," "empty-word configurations," "generalized Markov shift," and "continuous extension." It found no later paper giving an infinite-E_A example. A November 2025 author presentation still displayed the finite periodic-renewal family. The novelty verdict is therefore *apparently new within the bounded search*, not an exhaustive bibliographic guarantee.

Human review should focus on the standard column-limit/empty-root correspondence and on the one-letter marker-stem case in the last paragraph of the proof. Subject to those checks, the packet should be accepted as a full affirmative construction.

References

[1] R. Bissacot, I. Diaz-Granados, and T. Raszeja, *Extendable Shift Maps and Weighted Endomorphisms on Generalized Countable Markov Shifts*, arXiv:2506.07487 (2025).

[2] R. Bissacot, R. Exel, R. Frausino, and T. Raszeja, *Thermodynamic formalism for generalized Markov shifts on infinitely many states*, arXiv:1808.00765 (version cited by [1]).